
LET'S MAKE A
GAME
START

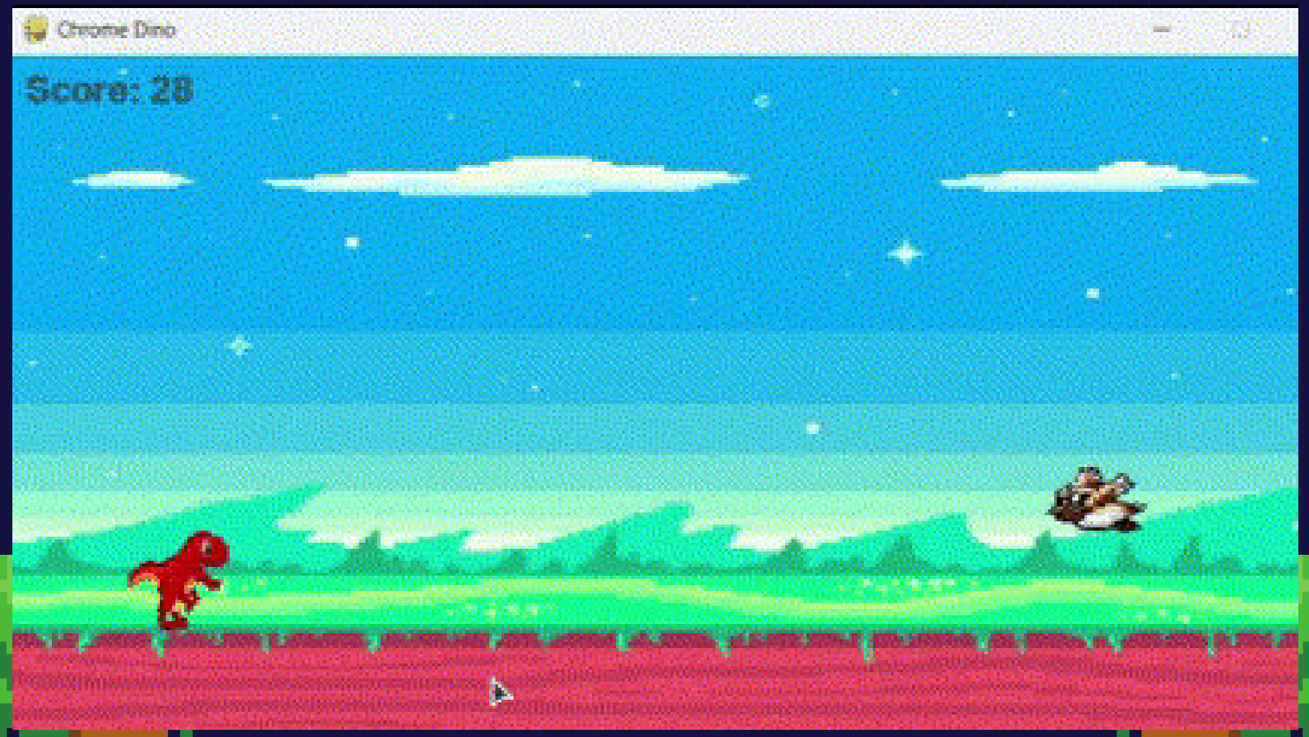
WELCOME TO OUR PYGAME WORKSHOP



We'll cover:

- Python Basics
- Basics of how a game is made
- Pygame basics
- Finally making an actual game

PREVIEW



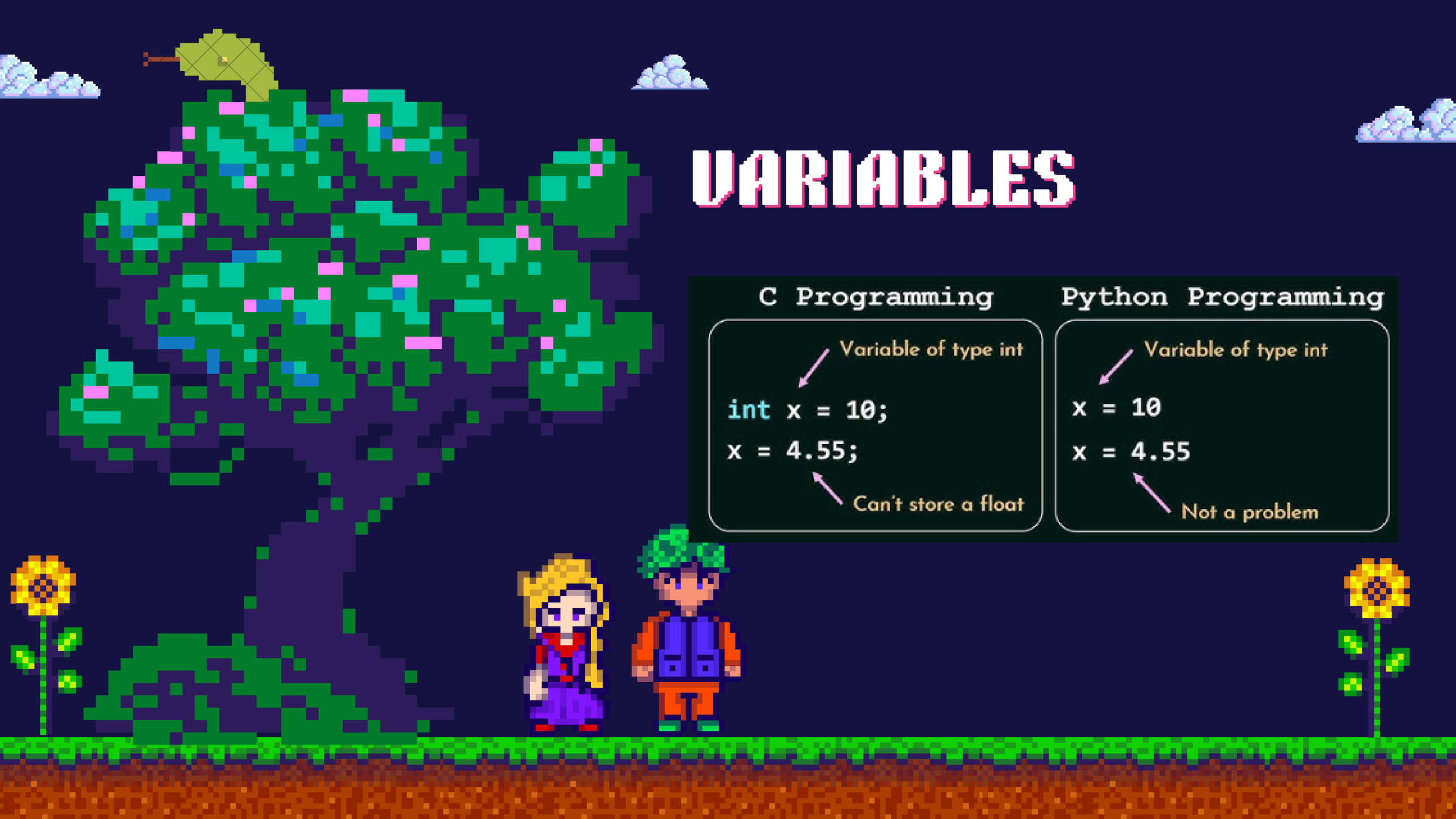

PYTHON

PROGRAMMING BASICS

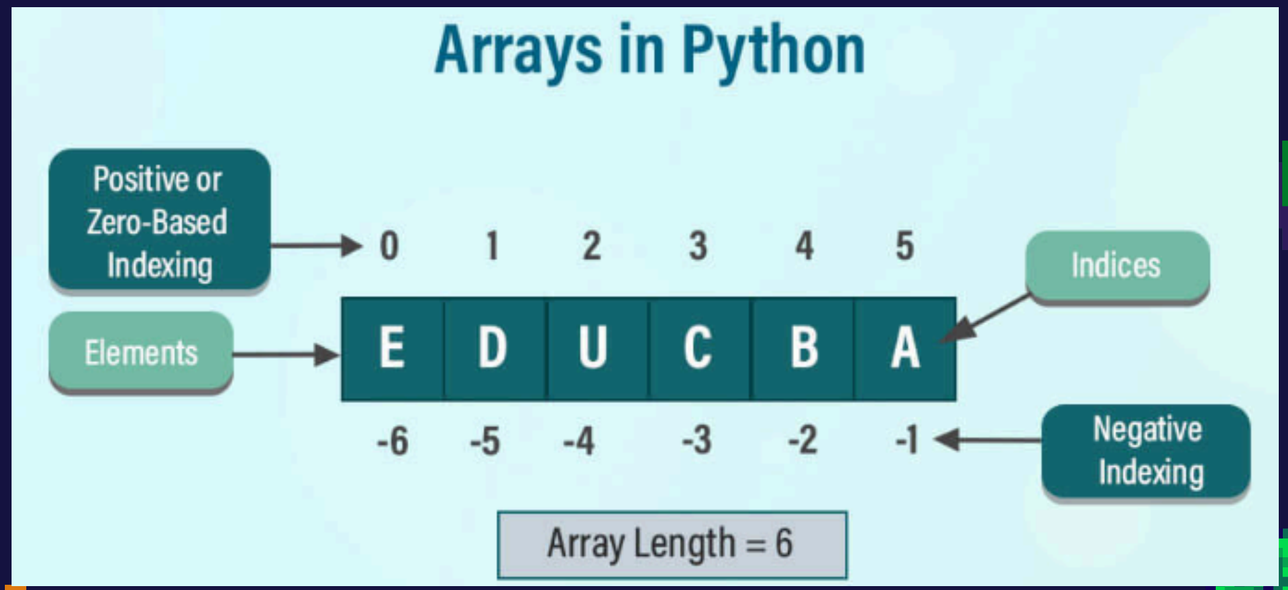
- Variables: Store data values using identifiers.
- Data Types: Examples - int, float, string, list, dict.
- Arrays (Lists): Ordered collection of elements.
- Dictionaries: Key-value pairs for structured data.
- Functions: Reusable blocks of code, defined using def.
- Loops: Automate repetitive tasks using for and while.



VARIABLES


C Programming
Variable of type int
int x = 10;
x = 4.55;
Can't store a float
Python Programming
Variable of type int
x = 10
x = 4.55
Not a problem

ARRAYS (LISTS)



DICTIONARY



```
a = {
    "name": "Dino",
    "age": 20
}

print(a["name"]) # Dino
```

LOOPS

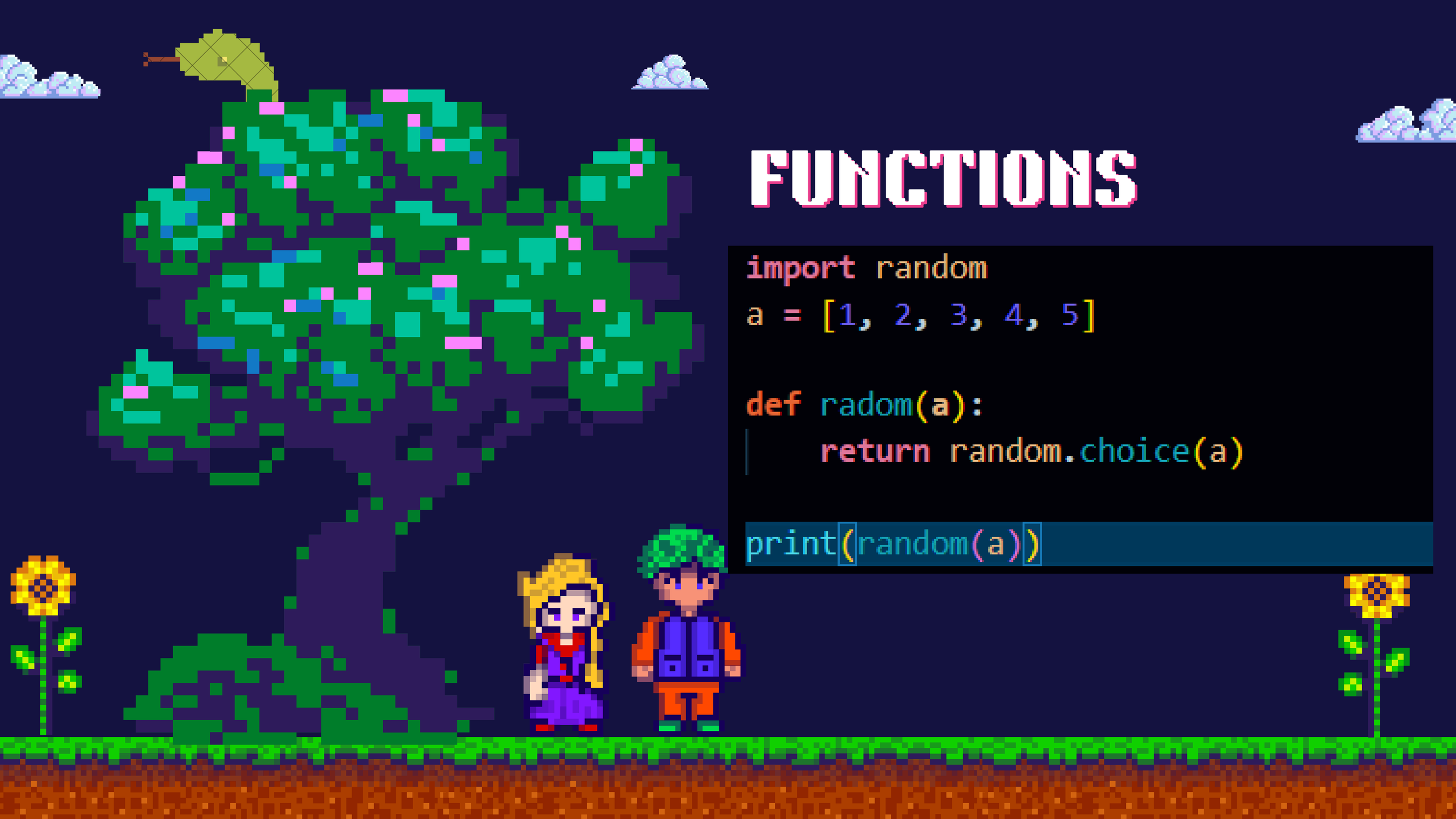
```python
a = [1, 2, 3, 4, 5]

for i in range(0, len(a)):
    print(a[i])

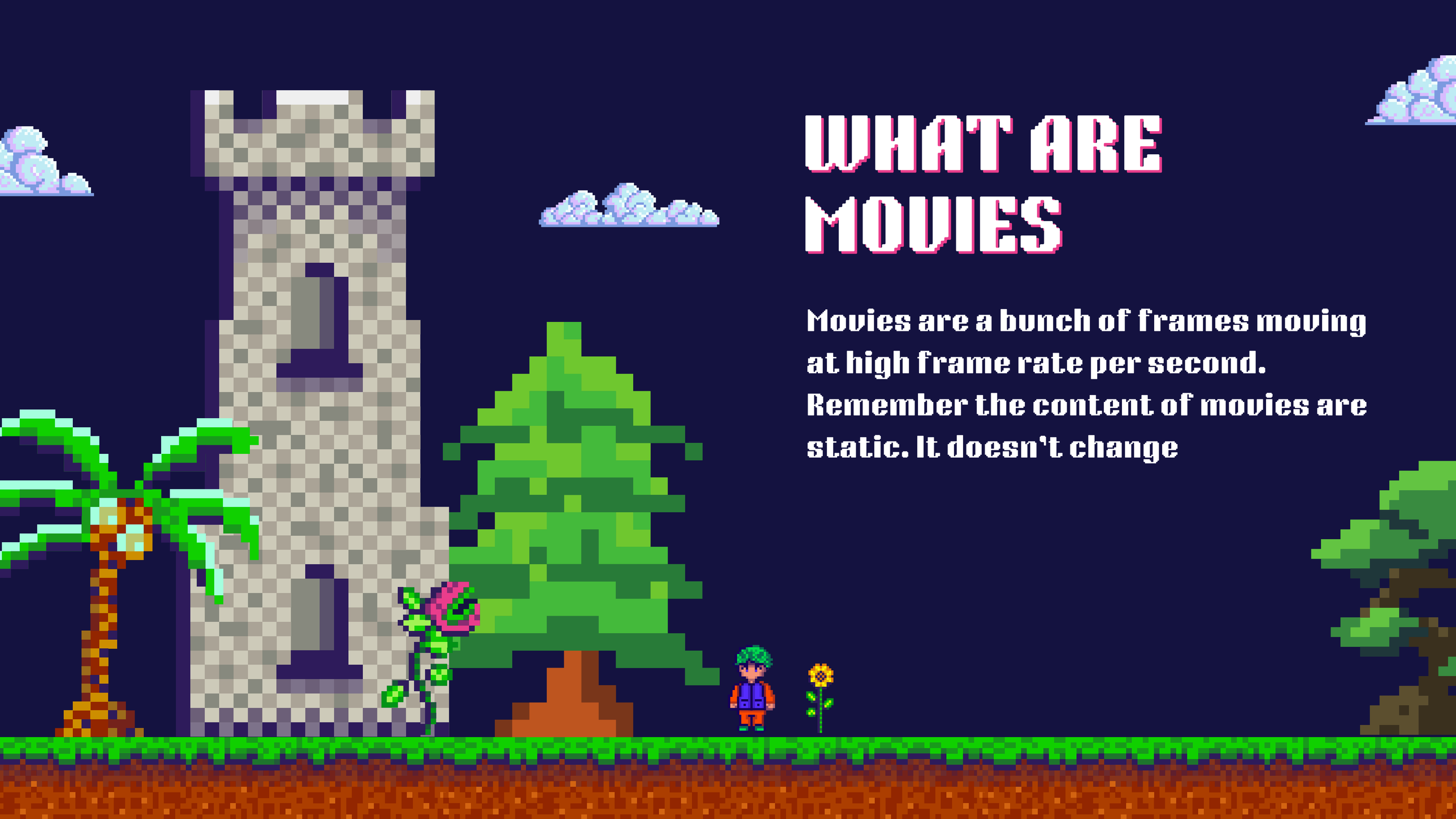


i = 0
while i < len(a):
    print(a[i])
    i += 1
```




FUNCTIONS
import random
a = [1, 2, 3, 4, 5]
def radom(a):
return random.choice(a)
print(random(a))


VIDEO
GAMES


WHAT ARE MOVIES
Movies are a bunch of frames moving at high frame rate per second. Remember the content of movies are static. It doesn't change

MOVIES VS GAMES

Games are also a bunch of frames moving at high frame rates. The main difference between a movie and a game lies in its content. In games content changes dynamically according to player's input.




HOW DOES
PYGAME WORKS?
Pygame keeps checking for player inputs and paints objects on screen accordingly. And it does this multiple times per second usually 30-60 times.

YOU CAN'T MAKE AAA GAMES WITH PYGAME

To make games like GOW, GTA V you need gane engines like unity or unreal engine which has bunch of inbuilt tools which pygame doesn't have!

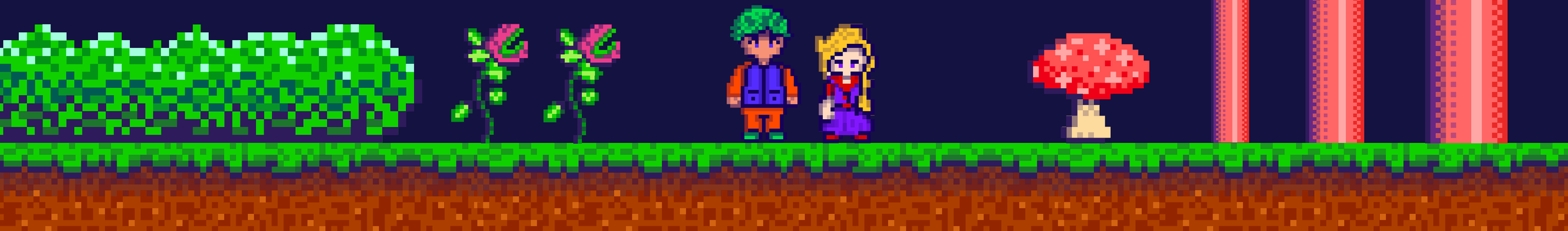



WHY PYGAME?
• It doesn't provide any inbuilt stuff so you have to build everything from scratch which makes you a good programmer
• Its easy to learn and really beginner friendly good for making cool stuff for beginners

SETUP

01 Install Python from the original website

02 Type pip install pygame in terminal



PYGAME BASICS

Functions	Description
pygame.init() pygame.mixer.init()	Initializes pygame modules
pygame.display.set_mode((100, 200))	sets the size of screen and returns a screen object
clock = pygame.time.Clock() clock.tick(60)	Sets the frame rate of game
pygame.event.get()	get different events of game
screen.blit(BG, (0, 0))	Displays image onto the screen

RECTANGLE OBJECT

A super useful object that helps to position images, check for collisions etc.

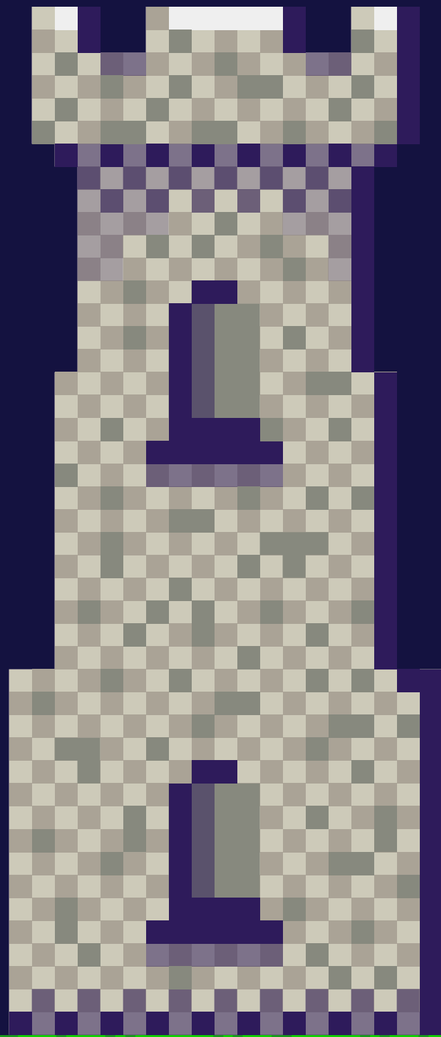
Functions	Description
DINO.get_rect(midbottom=(100, GROUND_HEIGHT))	Get rectangular object from image and position it
dino_rect.y dino_rect.x	access coordinates of rectangle object
dino_rect.colliderect(another_rect)	Check for collisions


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GAME

WE ENCOURAGE YOU TO READ DOCS AND EXPLORE MORE







REFERENCES



https://www.pygame.org/docs/

https://github.com/Rishab87/Onid


THANK
YOU